

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper No. BPT-501: Mathematical Physics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **five** of the following:

[5 × 4 = 20]

- (a) Find the magnetic induction $\vec{B}$ (in spherical polar coordinates), given the magnetic vector potential $\vec{A} = \frac{\vec{m} \times \vec{r}}{r^3} = \frac{m \sin \theta}{r^2} \hat{\phi}$, where $\vec{m} = m\hat{z}$ is a vector of constant magnitude pointing in the z -direction.
- (b) What are covariant and contravariant vectors? Explain each with a physical example.
- (c) If A is a nonsingular symmetric matrix, show that A^2 is also symmetric.
- (d) Evaluate $\nabla^2(1/r)$. What happens at the origin? Explain.
- (e) Find the Fourier series of the function $f(x) = \sin^2(x)$ in the interval $-\pi \leq x \leq \pi$.
- (f) Normalize the Laguerre polynomial $L_2(x) = 1 - 2x + \frac{x^2}{2}$ in the range $[0, \infty)$ with weight function e^{-x} . Construct a second-order, ordinary, linear, homogeneous differential equation (2-ODE) that is satisfied by this polynomial.

Question 2.

- (a) Verify the following tensor relations in $\mathbb{R}^3$ using Levi-Civita symbols and Kronecker deltas:

(i) $\epsilon_{ipq}\epsilon_{ipq} = 6$

(ii) $\epsilon_{ijk}\epsilon_{pqk} = \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}$

[6]

- (b) Write down the spherical polar unit vectors in terms of Cartesian unit vectors. Using these, calculate the partial derivatives of $\hat{r}, \hat{\theta}, \hat{\phi}$ with respect to r, θ, ϕ .

[8]

OR

- (a) Determine the circular cylindrical coordinate components of velocity and acceleration of a moving particle with position vector $\vec{r}(t)$.

[8]

- (b) Given a vector in spherical polar coordinates:

$$\vec{F} = \frac{2P \cos \theta}{r^3} \hat{r} + \frac{P \sin \theta}{r^3} \hat{\theta}$$

express it in Cartesian coordinates. If $\vec{F}$ represents a force field, does a potential function exist? If so, find it. Describe a physical system where this field arises.

[6]

Question 3.

- (a) Given that $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$ is one solution of the Bessel equation of order $1/2$, find a second linearly independent solution (up to a constant multiplier). [6]

- (b) Given the generating function $g(x, t) = e^{2xt - t^2} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n$ for Hermite polynomials, show that:

$$\int_{-\infty}^{\infty} e^{-x^2} [H_n(x)]^2 dx = 2^n n! \sqrt{\pi}$$

[8]

OR

- (a) Using the generating function $g(x, t) = (1 - 2xt + t^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(x) t^n$, ($|t| < 1$) for Legendre polynomials, show that:

$$P'_{n+1}(x) - P'_{n-1}(x) = (2n+1)P_n(x)$$

where the prime symbol ' denotes the derivative with respect to x . [6]

- (b) Solve Legendre's differential equation $(1 - x^2)y'' - 2xy' + n(n+1)y = 0$ using the Frobenius series method about the ordinary point $x = 0$. Show that when n is an integer, one of the solutions terminates to form a polynomial. [8]

Question 4.

- (a) By expressing the vector cross products in terms of the Levi-Civita symbols (ϵ_{ijk}), verify the vector identity:

$$(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C})$$

[6]

- (b) Find the eigenvalues and normalized eigenvectors of the matrix A :

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -\sqrt{2} \\ 0 & -\sqrt{2} & 0 \end{pmatrix}$$

How are these related to the eigenvalues and eigenvectors of A^{-1} ? [8]

Question 5.

- (a) Sketch a periodic square wave function $f(x)$ with period $T = 2$ and amplitude $A = 1$ which is also an even function (such that $f(x) = 1$ for $-0.5 < x < 0.5$, and $f(x) = 0$ for $0.5 < |x| < 1.0$). Find the Fourier series for this periodic function. [8]

- (b) The transformation relation between the unit vectors in circular cylindrical coordinates and Cartesian coordinates can be written as:

$$\begin{pmatrix} \hat{\rho} \\ \hat{\phi} \\ \hat{z} \end{pmatrix} = C \begin{pmatrix} \hat{i} \\ \hat{j} \\ \hat{k} \end{pmatrix}$$

Write down the matrix C explicitly and prove that it is orthogonal.

[6]